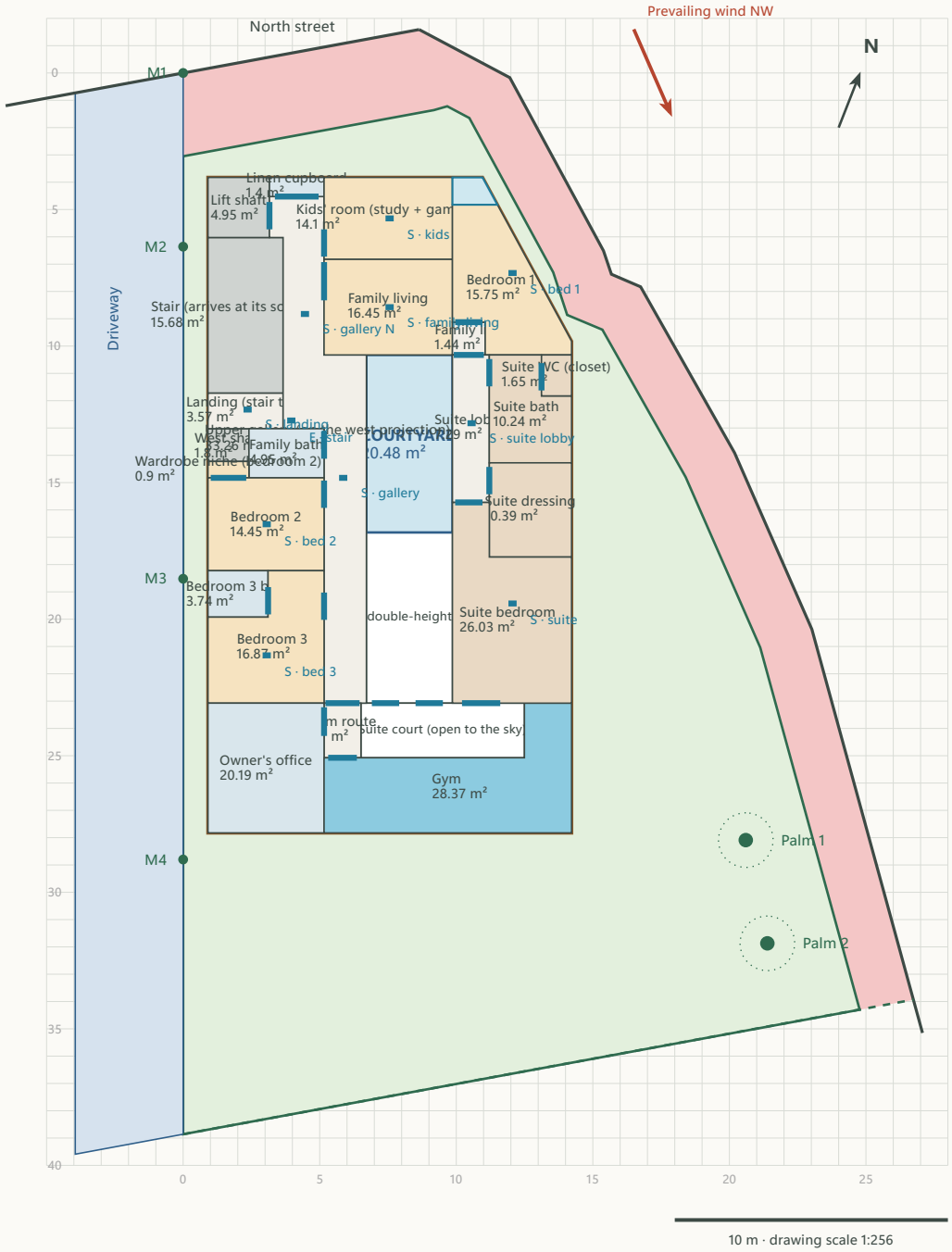


Taif / Palm House

First floor V7.1 · family part + the L (office, gym, suite court, master suite)

Taif / Palm House V7.1 · rooms in data/rooms_ff.csv · design and furniture in studies/ff_family.typ and studies/ff_suite.typ · schedule, light check and the split projection on the next pages

First floor V7.1. The gallery runs on a **1.55 m west projection** along the courtyard and the double-height void (x 4.25–5.8, y 6.5–19.25), on an **edge beam and 3 columns** at x 5.8, each in an existing wall or glass line; the first-floor west wall stands on the GF west wall (0.785 m from the M line). Courtyard sky and void 3.15 m on axes. West rooms 4.25 m and gallery 1.55 m on axes: **bedroom 2**, **bedroom 3 with its own bath**, the **family bath** by the west shaft. North band: **kids' room (study + gaming)** and the **family living** on the courtyard edge. **Bedroom 1** is one room on the straight y 6.5 line. The **L**: **owner's office** (west), **gym route**, **L-shaped gym** (the treadmill alone in its east arm, an acoustic double wall to the suite bedroom), the **suite court** open to the sky in front of the clerestory windows, and the **master suite** (lobby, luxury bath with WC closet, walk-in dressing, bedroom). No east duct. Stair, lift, landing and west shaft rise from the ground floor. Colours: suite (beige) · family (yellow) · baths, office and linen (blue-grey) · gym (blue) · circulation (light) · fixed (grey) · courtyard, void and court (white / blue). Blue marks: doors and the clerestory windows. Areas: available 245.72 m² · rooms 233.75 m² (circulation 48.27 m²) · suite court 11.97 m².
✓ **data/rooms_ff.csv matches the two studies room for room.**

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**309.68 m²**
Other open voids	**31.66 m²**
Courtyard (open to sky)	**20.48 m²**
Rooms placed (23)	**256.18 m²**
Limit minus courtyard and voids	**257.55 m²**
Air channel + duct	**1.37 m²**
Not yet assigned	**1.37 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Stair (arrives at its south end)	M	2.75 × 5.7	15.68 m²
Lift shaft	M	2.25 × 2.2	4.95 m²
West shaft	M	1.5 × 1.2	1.8 m²
Landing (stair top)	M	2.75 × 1.3	3.57 m²
Upper gallery (on the west projection)	M	3.55 × 18.55	33.26 m²
Linen cupboard	M	2 × 0.7	1.4 m²
Family bath	M	2.75 × 1.8	4.95 m²
Wardrobe niche (bedroom 2)	M	1.5 × 0.6	0.9 m²
Bedroom 2	M	4.25 × 3.4	14.45 m²
Bedroom 3 bath	M	2.2 × 1.7	3.74 m²
Bedroom 3	M	4.25 × 4.85	16.87 m²
Kids' room (study + gaming)	M	4.7 × 3	14.1 m²
Family living	M	4.7 × 3.5	16.45 m²
Family link	M	1.2 × 1.2	1.44 m²
Bedroom 1	M	4.361 × 5.5	15.75 m²
Gym route	M	1.35 × 2	2.7 m²
Owner's office	M	4.25 × 4.75	20.19 m²
Gym	M	9.061 × 4.75	28.37 m²
Suite lobby	M	1.35 × 5.4	7.29 m²
Suite bath	M	3.011 × 3.95	10.24 m²
Suite WC (closet)	M	1.1 × 1.5	1.65 m²
Suite dressing	M	3.011 × 3.45	10.39 m²
Suite bedroom	M	4.361 × 7.35	26.03 m²

First floor V6.3 · room schedule

wall centre lines (axes) · clear ≈ axes minus half a wall on every edge (from the studies) · CSV coordinates

Room	Part	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	family	2.75 × 1.3	3.57	3.14
Upper gallery (on the west projection)	family	polygon	33.26	30.08
Linen cupboard	family	2 × 0.7	1.4	–
Family bath	family	2.75 × 1.8	4.95	4.04
Wardrobe niche (bedroom 2)	family	1.5 × 0.6	0.9	–
Bedroom 2	family	4.25 × 3.4	14.45	12.84
Bedroom 3 bath	family	2.2 × 1.7	3.74	2.92
Bedroom 3	family	polygon	16.87	14.97
Kids' room (study + gaming)	family	4.7 × 3	14.1	12.44
Family living	family	4.7 × 3.5	16.45	15.16
Family link	family	1.2 × 1.2	1.44	1.08
Bedroom 1	family	polygon	15.75	13.83
Gym route	L	1.35 × 2	2.7	2.03
Owner's office	L	4.25 × 4.75	20.19	18.16
Gym	L	polygon	28.37	25.26
Suite court (open to the sky)	L	5.99 × 2	11.97	–
Suite lobby	L	1.35 × 5.4	7.29	5.94
Suite bath	L	polygon	10.24	8.79
Suite WC (closet)	L	1.1 × 1.5	1.65	1.09
Suite dressing	L	3.01 × 3.45	10.39	9.01
Suite bedroom	L	polygon	26.03	23.56
Rooms (indoor)			**233.75**	
Suite court (open)			**11.97**	
Stair, lift, west shaft, air tower			23.8	

Key numbers

- **Master suite:** 55.6 m² indoors (lobby, bath + WC closet, dressing, bedroom), + the suite court 11.97 m².
- **Circulation:** 48.27 m² (landing, gallery, family link, gym route, suite lobby).
- **Courtyard sky** 3.15 × 6.5 m and **double-height void** 3.15 × 6.25 m on axes; the ground-floor courtyard and living keep their full floor.
- **West rooms:** 4.25 m on axes (≈ 4.025 m clear); **gallery** 1.55 m (≈ 1.42 m clear); **family bath** 2.75 m wide.
- **Edge beam** on x 5.8 (spans 6.5 and 6.25 m) on 3 columns; the FF west wall on the GF west wall, 0.785 m from the M line.

Where the detail is

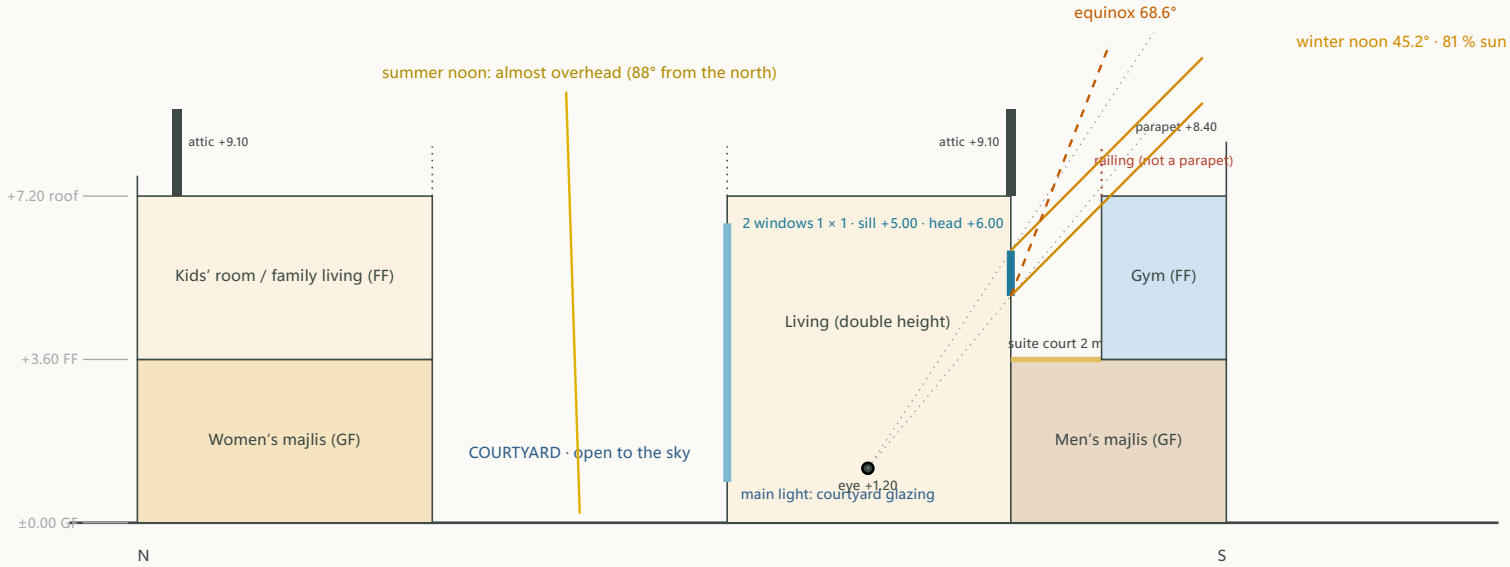
- **Furniture, doors, windows, drains, walking distances, routes ≥ 1.0 m and room checks:** studies/ff_family.typ (part 1) and studies/ff_suite.typ (part 2).
- **Decisions with reasons, construction notes, changelog:** docs/PROJECT_V6.md · **open items:** docs/OPEN_ITEMS.md.

Open items (first floor)

- Engineer: the edge beam (spans 6.5 and 6.25 m) and the 3 columns at x 5.8: sizes and foundations.
- Engineer: loads and vibration of the gym and the office over the men’s majlis; the y 3.0 partition over the women’s majlis.
- MEP: the stack outside the east wall (suite bath, vent stack), the kitchen extract through the east wall, the hood south of y ≈ 7.5.
- The clerestory sill against the suite court floor (eye height, or obscured glass); a daylight check of the courtyard.

First floor V7.1 · light check · the double-height living’s south windows

section north → south through the middle band (CSV x ≈ 7.3) · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · parapets 1.1



Behind the 2 m suite court	Winter noon sun on the windows	Equinox noon sun	Sky seen from the eye
gym roof edge with a 1.1 m parapet (top +8.30)	0 %	100 %	42 %
gym roof edge with a railing only (solid top +7.20) · **V6.2**	81 %	100 %	100 %

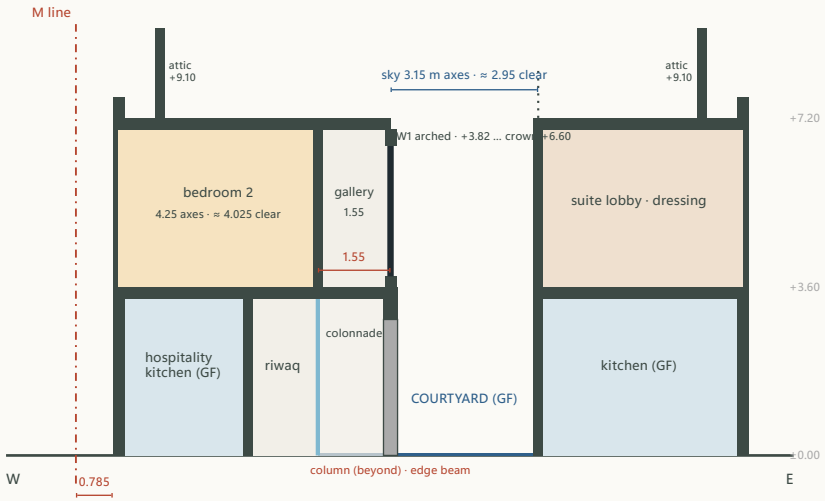
How the percentages are defined. “Sun on the windows”: the share of the window height (sill +5.00 to head +6.00) that receives direct sun at solar noon on 21 Dec (winter) or 21 Mar / 23 Sep (equinox) in this section; the ray from each point of the window must pass over the top of the gym’s north wall 2 m away. “Sky seen”: from a seated eye 1.2 m high in the double-height living (CSV y 16.1), the share of the windows’ vertical angle that shows sky rather than the gym. Pure geometry: no reflections, no sky model; a daylight simulation comes later.

- **The main light of the double-height living comes from the courtyard.** The 2 small south windows add light and let warm air out high on the south.
- **The 2 m in front of them is the suite court**, walled on four sides and open to the sky; the gym’s north wall is its south side, ending in a **railing, not a parapet**.
- **Privacy:** the windows open onto the suite court, so their sill stays above eye level from the court floor (or obscured glass); to check against the floor levels.
- **Summer:** the sun is almost overhead; light-coloured court walls and small hoods over the windows.

First floor V7.1 · the west projection on three columns

sections looking north, CSV x · the column on x 5.8 is beyond the cut (columns at y 6.5, 13, 19.25)

A–A · through the courtyard (y 12.0)



B–B · through the double-height living (y 16.5)



Item	Split projection (V5.1–V6.0, reverted)	V6.2 · 1.35 m inward on columns	Change
Courtyard sky opening, axes (≈ clear)	4.025 (3.825) m	3.15 (2.95) m	−0.875
Double-height void width, axes	4.025 m	3.15 m	−0.875
West room width, axes (≈ clear)	4.25 (4.025) m	4.25 (4.025) m	=
Gallery width, axes (≈ clear)	1.35 (1.225) m	1.55 (1.42) m	+0.2
Columns in the courtyard / living	none · 0.675 m cantilever	**3** on x 5.8 + edge beam	+3
FF west face → M line	0.11 m (approval needed)	**0.785 m** (on the GF wall)	+0.675 m
Walls on cantilever beams	FF west wall, 14.95 m	none	–
FF outline (axes)	318.86 m²	309.68 m²	−9.176 m²
Courtyard sky + double-height void	51.32 m²	40.16 m²	−11.158 m²
FF floor (outline − voids)	267.54 m²	269.52 m²	+1.981 m²
· family bath	3.74 m² (2.075 wide)	4.95 m² (2.75 wide)	+1.21 m²

V6.3 read from data/rooms_ff.csv; the split projection (V5.1–V6.0) from tag v6.0. Clear ≈ axes − ½ wall each side (250 outer, 200 inner, ≈ 50 mm for the steel window W1 on the void edge, V6.4); sky opening clear = axes − ½ of the east wall − the slab edge with its upstand.

- **Projection:** 1.55 m over the courtyard and the double-height void; the west rooms (4.25 m) and the gallery (1.55 m) take the band over the GF west band and riwaq.
- **Structure:** an edge beam on x 5.8 (spans 6.5 and 6.25 m) on 3 columns, each in an existing wall or glass line: the women's majlis south wall, the courtyard / living glass, the living's south wall. On the ground floor it reads as a colonnade along the courtyard. Walls stand on walls; nothing hangs on cantilever beams.
- **V6.1:** the split projection of V5.1–V6.0 (0.68 m in + 0.68 m out, the west wall on cantilever beams 0.11 m from the M line) is reverted: a wall on cantilever beams along ≈ 14.95 m costs more and is less direct than walls on walls, and the M-line projection needed municipal approval.
- **Cost of the revert:** courtyard sky 4.025 → 3.15 m; 3 columns on the courtyard edge.